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ScienceDirect
Building and Environment 43 (2008) 1–6
BUILDING AND ENVIRONMENT
www.elsevier.com/locate/buildenv

A multivariate-logistic model for acceptance of indoor environmental quality (IEQ) in offices

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Received 13 September 2006; received in revised form 9 January 2007; accepted 9 January 2007

Abstract

The indoor environmental quality (IEQ) in offices is examined from the prospect of an occupant's acceptance in four aspects: thermal comfort, indoor air quality, noise level and illumination level. Based on the evaluations made by 293 occupants of the IEQ of offices in Hong Kong, empirical expressions have been proposed to approximate an overall IEQ acceptance of an office environment at certain operative temperature (T_o), carbon dioxide concentration (CO_2), equivalent noise level (L_{eq}) and illumination level (lux). The overall IEQ acceptance is calculated from a multivariate logistic regression model. A range of acceptance in typical office environmental conditions and its dependence on the four parameters stated above are determined for design conditions. The proposed overall IEQ acceptance can be used as a quantitative assessment criterion for an office environment and similar environment where an occupant's evaluation is expected.

Keywords: Indoor environmental quality; Multivariate logistic regression; Field survey; Air-conditioned offices

1. Introduction

Today the concept of an acceptable indoor environmental quality (IEQ) as an integral part of the total building performance approach is still not fully appreciated. Physical environmental parameters such as air temperature, relative humidity, acoustics, air quality, lighting, ventilation and air distribution are all interrelated, and the feeling of comfort is a composite state of an occupant's mind responding to the senses to these factors [1–4]. This state of mind is an intricate response to the indoor environmental factor groups, including physical environment sustained by the building and its services system, and individual physiological conditions such as health, social relations, financial state, etc.

Studies showed that an occupant's acceptance of an environment depended on a number of environmental parameters [5]. Four basic components, namely thermal comfort, indoor air quality, aural and visual comfort were identified for determining an acceptable IEQ. Conventional studies on indoor environment address each of them separately. They are still addressed independently by designers for many office designs. More recently, the equivalence of the discomfort caused by different physical qualities has been considered [6–8]. Discomforts caused by indoor air pollution, thermal load and noise were investigated. It was reported that at an operative temperature between 23 and 29 °C, each degree Celsius change would associate the same effect on human comfort with a change in perceived air quality of 2.4 decipol, or a change in noise level of 3.9 dB [7]. For levels of perceived air quality up to 10 decipol, a unit change had the same effect on human comfort as a change in noise level of 1.2 dB [6]. The equivalence of acoustic sensation to the thermal one was proposed for short-term exposure that each degree Celsius change in temperature had the same effect of 2.6 dBA [8]. Workplace variables inducing the largest number of health symptoms, comfort or odour concerns were investigated by multivariate regression analysis [9–15]. It was realised that successful control of the indoor

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doi:10.1016/j.buildenv.2007.01.001

environment required an understanding of the integral indoor environmental parameters. An overall IEQ index would be derived to describe the state of mind of a user in balance with the indoor environment.

Unacceptable indoor environments are often manifest in some forms and symptoms of sick building syndrome (SBS) prevail in many office buildings [11]. This study argues that the subjective evaluation of an indoor environment being perceived by an occupant can be used to assess the acceptance of the IEQ. In particular, occupants' acceptance of the four basic components of IEQ was evaluated and correlated with the overall IEQ acceptance of an office environment. The occupants' attitudes towards the operative temperature, CO_2 concentration, equivalent noise level and illumination level and the overall IEQ acceptance recorded by a dichotomous scale were studied [5,16–18]. Mathematical expressions were proposed for the overall IEQ acceptance using a multivariate logistic regression model with the former four parameters recorded. The proposed overall IEQ acceptance can be used as a quantitative assessment criterion for an office environment and similar environment where an occupant's evaluation is expected.

2. Methodology

Subjective evaluations made by 293 occupants of indoor environmental conditions in typical air-conditioned offices in Hong Kong were studied. The sample offices had floor areas ranged from 90 to 1200 m^2 and covered Grades A–C offices in Hong Kong, where, *Grade A* offices were spacious and furnished with high-quality finishes, flexible layout, large floor plates, well decorated lobbies and circulation areas, effective central air-conditioning, good lift services zoned for passengers and goods deliveries, professional management, and parking facilities were normally available; *Grade B* offices were of ordinary design with good-quality finishes, flexible layout, average-sized floor plates, adequate lobbies, central or free-standing air-conditioning, adequate lift services, good management, and parking facilities were not essential; and *Grade C* offices were those plain with basic finishes, less flexible layout, small floor plates, basic lobbies, hardly any central air-conditioning, barely adequate or inadequate lift services, minimal to average management and no parking facilities.

The occupant's acceptance of the perceived indoor environment given by four aspects, namely thermal environment, indoor air quality, equivalent noise level and illumination level, was studied with a dichotomous assessment scale [19]. This scale was used for a direct feedback of acceptability with the question 'Is the thermal environment/indoor air quality/noise level/illumination level being perceived in the office environment acceptable to you?' being asked. The ranks '(1) Yes, acceptable' and '(0) No, not acceptable' were self-explanatory. In order to confirm the validity of their responses, each respondent had to use a semantic differential evaluation scale for the subjective assessment of the first two aspects, and a visual analogue assessment scale for the evaluation of the aural and visual comfort [19]. At the end of the survey, an overall acceptance of the IEQ was determined.

3. Overall acceptance of IEQ

For a typical local office environment being perceived by an occupant, this study suggests the overall IEQ acceptance θ is dependent on four environmental parameters, namely the operative temperature ζ_1(°C), the CO_2 concentration ζ_2 (ppm), the equivalent noise level ζ_3 (dBA) and the illumination level ζ_4 (lux),

$$\theta = f(\phi_i); \quad i = 1, \ldots, 4, \tag{1}$$

$$\phi_i = \phi_i(\zeta_i); \quad i = 1, \ldots, 4, \tag{2}$$

where ϕ_i is the level of acceptance of the four environmental factors.

The level of acceptance of a thermal environment ϕ_1 at certain operative temperature ζ_1 is related to the predicted percentage dissatisfaction (PPD) of thermal comfort [17,20],

$$\phi_1 = 1 - \frac{\text{PPD}}{100}. \tag{3}$$

From a recent study for 61 typical offices in Hong Kong, the percentage acceptance of indoor air quality ϕ_2, aural environment ϕ_3 and illumination level ϕ_4, given ζ_2, ζ_3 and ζ_4 at working plane, is correlated by Mui and Wong [21–23]:

$$\phi_2 = 1 - \frac{1}{2}\left(\frac{1}{1+\exp(3.118 - 0.00215\zeta_2)} - \frac{1}{1+\exp(3.230 - 0.00117\zeta_2)}\right); \quad 500 \leqslant \zeta_2 \leqslant 1800, \tag{4}$$

$$\phi_3 = 1 - \frac{1}{1+\exp(9.540 - 0.134\,\zeta_3)}; \quad 45 \leqslant \zeta_3 \leqslant 72, \tag{5}$$

$$\phi_4 = 1 - \frac{1}{1+\exp(-1.017 + 0.00558\zeta_4)}; \quad 200 \leqslant \zeta_4 \leqslant 1600. \tag{6}$$

4. Result and discussion

The surveyed office environmental conditions were typical, i.e. the operative temperature ζ_1, the carbon dioxide concentration ζ_2, the equivalent noise level ζ_3 and the illumination level ζ_4 at working plane were 18–25 °C, 500–1800 ppm, 45–72 dBA and 200–1600 lux, respectively [5].

A total of 293 occupants were interviewed and their evaluations of the IEQ and the four parameters were obtained. The results are summarized in Table 1. The correlation between subjective response to each parameter

Table 1
Occupant's votes on acceptance of a perceiving indoor environmental quality

Overall acceptance θ	Votes	Thermal environment ϕ_1		Air quality ϕ_2		Noise level ϕ_3		Illumination level ϕ_4	
		U	A	U	A	U	A	U	A
U	130	62	68	69	61	40	90	35	95
A	163	4	159	16	147	11	152	12	151
Total	293	66	227	85	208	51	242	47	246

U—unacceptable; A—acceptable.

and the overall IEQ acceptance was evaluated by a statistic χ^2-test. Results showed that all the four parameters contributed to the overall IEQ; and the significance between the acceptance votes on the latter and those on the former, ranking from the most important to the least, was as follows: $p = 3.3 \times 10^{-20}$ for thermal environment, $p = 5.2 \times 10^{-16}$ for air quality, $p = 7.1 \times 10^{-8}$for noise level and $p = 5.8 \times 10^{-6}$ for illumination level.

The overall IEQ acceptance θ for an office environment perceived by an occupant expressed by a multivariate logistic regression model is proposed,

$$\theta = 1 - \frac{1}{1 + \exp\left(k_0 + \sum_{i=1}^{4} k_i \phi_i(\zeta_i)\right)}, \tag{7}$$

where the regression constants determined from the 293 occupant evaluations are

$$k_i = \begin{cases} -15.02 \\ 6.09 \\ 4.88 \\ 4.74 \\ 3.70 \end{cases} ; \quad i = 0, \ldots, 4. \tag{8}$$

Values of k_1–k_4 confirm the relative importance of the four contributors to θ, the larger the value, the greater the importance. Indeed, in the surveyed offices, most of the reports on unsatisfactory environment were related to thermal comfort. Apparently, occupants were very sensitive to the operative temperature as compared with the other three parameters.

Various combinations of contributors $i = 1, \ldots, 4$ and the corresponding overall IEQ acceptance were considered. A total of $j = 2^4$ possibilities were found. Taking the binary notation for the acceptance, i.e. 0 and 1 for 'Unacceptable' and 'Acceptable', respectively, and ranking the four contributors in the order from most important to least as shown in Table 2, the surveyed overall IEQ acceptance θ_s of case j is expressed by

$$\theta_{s,j} = \left(\frac{N_{\phi=1}}{N}\right)_j; \quad j = 1, \ldots, 16, \tag{9}$$

where N is the survey sample size and $N_{\phi=1}$ is the acceptance count.

Table 2
Overall IEQ acceptance

Case No.	Survey sample N	Survey acceptance θ_s	Contributors				Predicted acceptance of IEQ, θ
			ϕ_1	ϕ_2	ϕ_3	ϕ_4	
1	0	—	0	0	0	0	2.99×10^{-7}
2	6	0	0	0	0	1	1.21×10^{-5}
3	1	0	0	0	1	0	3.44×10^{-5}
4	18	0	0	0	1	1	0.0014
5	4	0	0	1	0	0	3.9×10^{-5}
6	4	0	0	1	0	1	0.0016
7	7	0	0	1	1	0	0.0045
8	26	0.15	0	1	1	1	0.1500
9	1	0	1	0	0	0	0.0001
10	10	0.20	1	0	0	1	0.0053
11	12	0	1	0	1	0	0.0150
12	37	0.38	1	0	1	1	0.3792
13	4	0	1	1	0	0	0.0170
14	22	0.41	1	1	0	1	0.4108
15	18	0.67	1	1	1	0	0.6659
16	123	0.99	1	1	1	1	0.9877

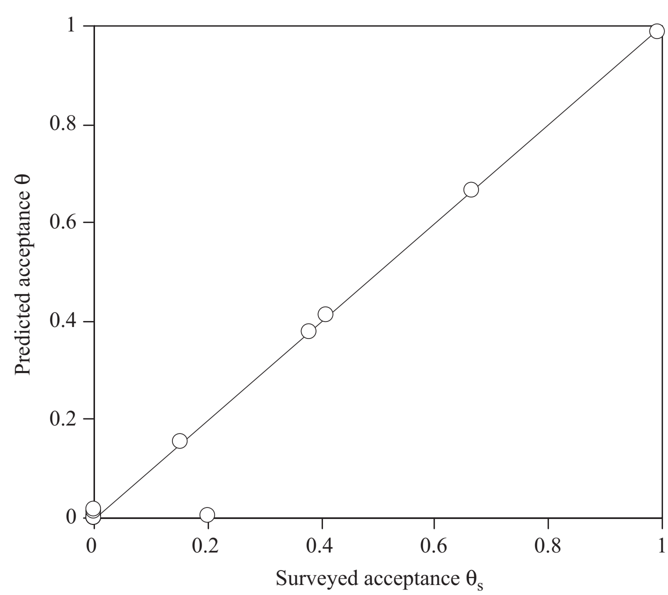

Fig. 1. Overall IEQ acceptance.

In the survey, the case of $j = 1$ was not recorded. A total of 7 cases had less than 10 records and only 2 cases had more than 30, as shown in the table. It was noted that the surveyed IEQ acceptance for those cases having a sample size less than 30 would be subject to considerable uncertainty. A good agreement was found between the predicted and the surveyed results for cases with a sample size more than 20. The predicted overall IEQ acceptance is plotted against the surveyed one as shown in Fig. 1. The sample correlation coefficient was 0.9856 ($p = 2 \times 10^{-11}$), which indicated a strong linear association between the predicted and surveyed overall IEQ acceptance.

The overall IEQ acceptance for offices due to the four contributors ζ_1, ζ_2, ζ_3 and ζ_4 was examined via the proposed model in Eq. (7). Fig. 2 shows the dependence of the predicted overall IEQ acceptance on the variations of any two contributors, while the other two remaining unchanged under the nominal conditions. The nominal conditions selected in this study were the typical design values for offices, i.e. $\zeta_1 = 25\,°C$, $\zeta_2 = 1000\,ppm$, $\zeta_3 = 57.5\,dBA$ and $\zeta_4 = 500\,lux$. As expected, Fig. 2 shows that the predicted overall IEQ acceptance is very sensitive to the operative temperature ζ_1 but less to the CO_2 concentration ζ_2; this acceptance drops gradually as noise level increases. A significant drop of the acceptance was found when the noise level was higher than 60 dBA. It was reported that the acceptance was relatively steady at an illumination level above 500 lux.

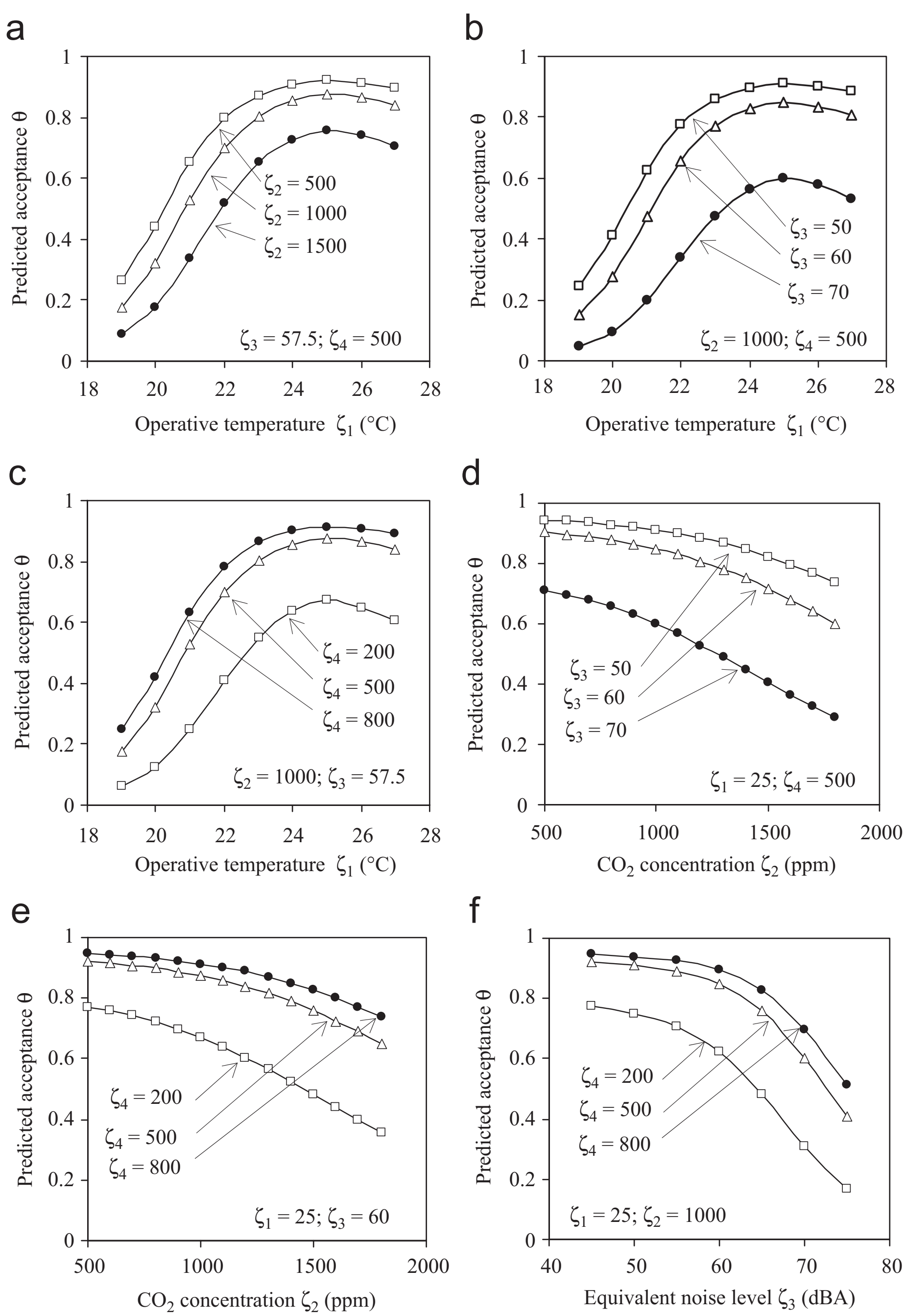

Fig. 2. Dependence of contributors in predicted IEQ acceptance for offices.

In order to illustrate the overall IEQ acceptance in typical offices, the environmental conditions in the surveyed offices is standardised by the CO_2 concentration ζ_2, the equivalent noise level ζ_3, and the illumination level ζ_4,

$$\Re = \int_{-\infty}^{\zeta_i} \frac{e^{-(\zeta_i-\mu_i)^2/2\sigma_i^2}}{\sigma_i\sqrt{2\pi}}; \quad i = 2,3,4, \tag{10}$$

where $\Re$ is the benchmark of the office environment that: 0% represents an environment having the highest ζ_2 and ζ_3, and lowest ζ_4, and 100% an environment having the lowest ζ_2 and ζ_3, and highest ζ_4; μ_i and σ_i are the mean and standard deviations of parameter ζ_i. It was reported that in the surveyed offices μ_i were 23.6 °C, 997 ppm, 56 dBA and 655 lux; and σ_i were 1.1 °C, 321 ppm, 3.8 dBA and 299 lux, respectively.

The predicted overall IEQ acceptance θ for the benchmark office environmental conditions $\Re = 0.05$–0.95 is shown in Fig. 3. At 24–26 °C, the typical operative temperature range in local offices, θ is above 90% for offices of benchmark 0.9 and above 80% for those of 0.5, but is only about 40% for offices of benchmark 0.1. The model implicitly shows that it is statistically impossible to have 0% dissatisfaction, from the data obtained in this study. From the observation of the predicted acceptance found in this study, together with the discussion from experienced professionals, the IEQ of offices can be classified into three levels according to the satisfaction rating, which in turn can be considered in the air-conditioning offices in Hong Kong:

(i) Good IEQ ($\theta \geqslant 90\%$)—It represents a Grade A office which has an indoor environmental performance better than average. It requires more attendance to achieve compliance with relevant standards for each of the basic IEQ components. It also requires more effort in design, installation supervision, high precision testing and commissioning, user-oriented operation and predictive maintenance. The project team should realise that more resources are required during construction, operation and maintenance. However, the return of higher satisfaction rate will reduce trouble-shooting and remedial costs. Besides, the systems may not require more energy to operate.

(ii) Average IEQ ($80\% \leqslant \theta < 90\%$)—It represents an average grade office. It can normally be achieved by traditional competence and ethical job when the appropriate standards for each of the basic IEQ components are complied with.

(iii) Bad IEQ ($\theta \leqslant 40\%$)—This represents a below average grade office which is often the result of an investment plan based on speculation. In the building services industry, SBS is said to occur if more complains are reported about the indoor environment. A scheme to improve the environmental control systems is urgently required in order to reduce problems. There may be some savings on initial cost and energy conservation but the low return of satisfaction may prove it to be not worthwhile. Since it is on the brink of being a problem building, the IEQ satisfaction should be specified.

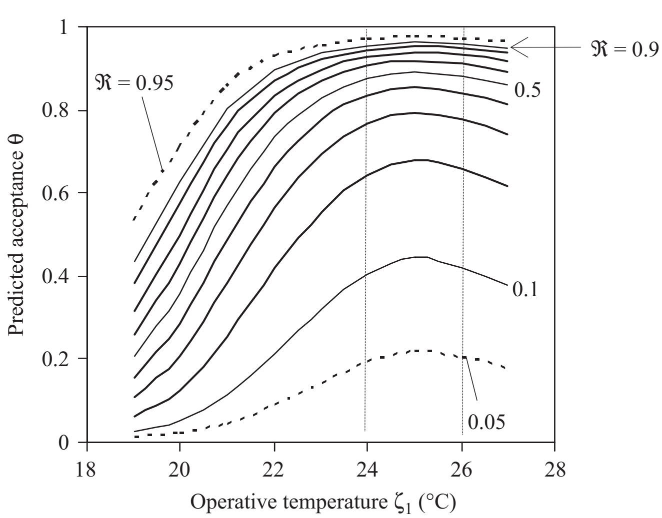

Fig. 3. Predicted IEQ acceptance for offices.

5. Conclusion

Office environment in Hong Kong is generally designed to the design guides and practices for the occupant's comfort. Until now, overall indoor environmental quality (IEQ) in offices in terms of an occupant's acceptance has not been considered in some local designs. In this work, the overall IEQ of offices in Hong Kong was evaluated by 293 occupants in four aspects, namely thermal comfort, indoor air quality, equivalent noise level and illumination level. The results showed that the operative temperature, carbon dioxide concentration, equivalent noise level and illumination level all had important effects on the overall IEQ acceptance. The relative significance of them, ranking from the most important to the least, was the indoor thermal environment, the air quality, the noise level and the illumination level. Empirical expressions have been proposed to approximate the occupant's acceptance of the IEQ in an office. At 24–26 °C, the typical operative temperature range in local offices, the predicted IEQ acceptance is above 80% for an office of good or average IEQ, but the acceptance will drop to below 40% for an office having a bad environment. From the observation of the predicted overall IEQ acceptance together with the decisions from experienced professionals, "good", "average" and "bad" levels of satisfaction can be considered in the air-conditioning offices in Hong Kong.

Acknowledgement

This work was undertaken in the Department of Building Services Engineering, The Hong Kong Polytechnic

University. Funding has been received from the Research Grants Council (Project No. PolyU5248/06E) and The Hong Kong Polytechnic University (GYE80).

Appendix A

The predicted percentage dissatisfied (PPD) at a predicted mean vote (PMV) for thermal comfort is determined by [17,20]

$$\mathrm{PPD} = 100 - 95 \times \mathrm{e}^{-(0.03353\ \mathrm{PMV}^4 + 0.2179\ \mathrm{PMV}^2)}; \quad -2 \leqslant \mathrm{PMV} \leqslant 2. \tag{11}$$

The PMV is given by

$$\begin{aligned}\mathrm{PMV} = {} & \left(0.303e^{-0.036\frac{M}{A_{\mathrm{Du}}}} + 0.028\right)\left\{\left(\frac{M}{A_{\mathrm{Du}}} - E_{\mathrm{W}}\right) - 3.05 \times 10^{-3}\right.\\ & \times \left[5733 - 6.99\left(\frac{M}{A_{\mathrm{Du}}} - E_{\mathrm{W}}\right) - p_{\mathrm{a}}\right] - 0.42\\ & \times \left[\left(\frac{M}{A_{\mathrm{Du}}} - E_{\mathrm{W}}\right) - 58.15\right]\\ & - 1.7 \times 10^{-5}\frac{M}{A_{\mathrm{Du}}}(5867 - P_{\mathrm{a}}) - 0.0014\frac{M}{A_{\mathrm{Du}}}(34 - T_{\mathrm{a}})\\ & - 3.96 \times 10^{-8} f_{\mathrm{cl}}[(T_{\mathrm{cl}} + 273)^4 - (T_{\mathrm{mrt}} + 273)^4]\\ & \left. - f_{\mathrm{cl}} h_{\mathrm{conv}}(T_{\mathrm{cl}} - T_{\mathrm{a}})\right\}.\end{aligned} \tag{12}$$

The surface temperature of clothing T_{cl} (°C) is given by

$$\begin{aligned}T_{\mathrm{cl}} = {} & 35.7 - 0.028\left(\frac{M}{A_{\mathrm{Du}}} - E_{\mathrm{w}}\right) - I_{\mathrm{cl}}\{3.96 \times 10^{-8} f_{\mathrm{cl}}\\ & \times [(T_{\mathrm{cl}} + 273)^4 - (T_{\mathrm{mrt}} + 273)^4] + f_{\mathrm{cl}} h_{\mathrm{conv}}(T_{\mathrm{cl}} - T_{\mathrm{a}})\}.\end{aligned} \tag{13}$$

The convective heat transfer coefficient h_{conv} ($\mathrm{W\,m^{-2}\,°C}$) is given by

$$h_{conv} = \begin{cases} 2.38(T_{\mathrm{cl}} - T_{\mathrm{a}})^{0.25} \\ 12.1\sqrt{U_{\mathrm{ar}}} \end{cases}; \quad \begin{cases} 2.38(T_{\mathrm{cl}} - T_{\mathrm{a}})^{0.25} \geqslant 12.1\sqrt{U_{\mathrm{ar}}} \\ 2.38(T_{\mathrm{cl}} - T_{\mathrm{a}})^{0.25} < 12.1\sqrt{U_{\mathrm{ar}}} \end{cases}. \tag{14}$$

The ratio of a man's surface area while clothed to a man's surface area while nude f_{cl} is given by,

$$f_{\mathrm{cl}} = \begin{cases} 1.00 + 1.290 I_{\mathrm{cl}}^{-1} \\ 1.05 + 0.645 I_{\mathrm{cl}} \end{cases}; \quad \begin{cases} I_{\mathrm{cl}} \leqslant 0.078 \\ I_{\mathrm{cl}} > 0.078 \end{cases}, \tag{15}$$

where A_{Du}($\mathrm{m^2}$) is the body surface area of a human; E_{W} ($\mathrm{W\,m^{-2}}$) the external work for an activity; I_{cl} ($\mathrm{m^2\,°C\,W^{-1}}$) is the thermal resistance of clothing; M(W) is the metabolic rate; P_{a} (Pa) is the partial water vapour pressure; T_{a}(°C) and T_{mrt}(°C) are the air temperature and the mean radian temperature; and U_{ar}($\mathrm{m\,s^{-1}}$) is the air velocity to the human body. Note that I_{cl} ($\mathrm{m^2\,°C\,W^{-1}}$) and f_{cl} are functions of the clothing; M/A_{Du} and E_{W} are functions of the activity.

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